

Module 13 – Estimating Proportions and Variances Handout

1. In considering the production of a new car accessory, a manufacturer wants to do a marketing study that includes determination of the proportion of cars with Apple Car Play. How many cars must be sampled if the manufacturer wants 92% confidence that the sample proportion is in error by no more than 0.035?
2. You want to estimate the percentage of employees who have been with their current employer for one year or less. Data from the U.S. Department of Labor suggest that this percentage is around 29%, but you must conduct a survey to be 99% confident that your randomly selected sample of employees leads to a sample percentage that is off by no more than two percentage points. How many employees must you survey?
3. A sports equipment supplier finds that when 1180 adults are surveyed, 79 play golf (based on data from a Roper Organization Poll). Construct the 95% interval estimate for the true proportion of all adults who play golf.
4. A survey of 1002 voters by Martilla and Kiley, Inc., showed that 13% of them felt that it is very likely that we will get into a nuclear war within the next 10 years. Construct the 94% confidence interval for the true percentage of all voters who feel that way.
5. A candy manufacturer selects mints at random from the production line and weighs them. For one week, the day shift weighed 194 mints, and the night shift weighed 162 mints. The numbers of these mints that weighed at most 21 grams was 28 for the day shift and 11 for the night shift. Let p_1 and p_2 denote the proportions of mints that weigh at most 21 grams for the day and night shifts, respectively. Construct a 96% confidence interval for the difference in the proportions of mints that weigh at most 21 grams, $p_1 - p_2$.
6. In an Airport Transit Association poll of 4664 adults, 3358 indicated that they have flown in an airplane. Find the 99% confidence interval for the percentage of all adults who have flown in an airplane.
7. For developing countries in Asia (excluding China) and Africa, let p_1 and p_2 be the respective proportions of preschool children with chronic malnutrition (stunting). If respective random samples of 1300 and 1100 yielded 520 and 385 children with chronic malnutrition, find an approximate 90% confidence interval for the difference in proportions, $p_1 - p_2$.
8. Consider a random sample of 45 turtles whose population of weights follows a normal distribution. The sample data shows a mean weight of 12.13 pounds and variance of 3.46 pounds. Find a 90% confidence interval for the population variance of this type of turtle.

9. Consider a normal distribution of final exam grades for a statistics course. A random sample of 34 statistics students shows a mean of 74 and standard deviation of 2.634. Find a 99% confidence interval for the population standard deviation for this statistics course final exam.

10. A random sample of 40 female employees of Moses Enterprises looking for employee satisfaction ratings on a scale of 1 to 5 shows a sample mean 4.18 with a sample standard deviation of 0.24. A random sample of 30 male employees of Moses Enterprises looking for employee satisfaction ratings on a scale of 1 to 5 shows a sample mean of 3.89 with a sample standard deviation of 0.59. Find a 95% confidence interval for the ratio of variances for female to male employees, assuming both distributions are normal.